MIAE 215 — Official 1-Page Master Cheat Sheet **Concordia · Gina Cody School** **Thu Oct 22 · 8:45–10 AM**
1 Side Only · 10pt Font · Only Aid Allowed · 40% Output Tracing / 60% Code

1. Data Types, Limits & #1 Division Trap

Type	Bytes	Range / Behavior
`int`	4	-2,147,483,648 to 2,147,483,647
`unsigned`	4	0 to 4,294,967,295 (wraps at 0)
`char`	1	`'0'`=48, `'A'`=65, `'a'`=97 (single quotes)
`bool`	1	`true`(1), `false`(0); non-zero is true
`double`	8	~15 digits; compare: `fabs(a-b)<1e-7`

- **Integer Division Truncates:** `7/2 = 3`, `1/2 = 0`, `-7/2 = -3`.
- **Fix:** Cast *before*: `(double)a/b`. **(double)(a/b)=3.0** (too late!).
- **Modulo %:** Ints only. Sign follows left: `-7%2 = -1`, `7%-2 = 1`.
- **Overflow:** `100000 * 100000` overflows! Use `100000LL * 100000`.

2. Operator Precedence & Short-Circuit

- **Order:** `()` → `! ++ -- (type)` → `* / %` → `+ -` → `< <= > >=` → `== !=` → `&&` → `||` → `= += -= *= /=`
- `a++` uses then increments; `++a` increments first.
- **Assignment in if:** `if(x = 5)` sets x=5, always true! Write `x == 5`.
- **Chained relational:** `1 < x < 5` → `(1<x) < 5` is always true! Must write `(1 < x) && (x < 5)`.
- **Short-circuit:** `A && B` skips B if A is false; `A || B` skips B if A is true.

3. Control Flow & Loops

- **if-else:** `else` binds to closest unmatched `if`.
- **Semicolon trap:** `for(i=0; i<5; i++);` → empty body runs 5 times!
- **Loops:** `for(init; cond; step)` (0+); `while(cond)` (0+); `do{...}while(cond);` (runs ≥ 1 time!).
- **Trips:** `i=0; i<10; i++` (10); `i=0; i<=10; i++` (11); `i=1; i<10; i++` (9).
- `break` exits loop/switch; `continue` jumps to update/next trip.
- **switch(x):** x must be integral (int/char/bool); each case needs `break;`!

4. Write-a-Program Accumulator Recipes (60%)

- **Sum:** `int s = 0; s += val;` (init with 0)
- **Product/Factorial:** `long long p = 1; p *= i;` (init with 1!)
- **Count / Filter:** `int c = 0; if (val > 0) c++;`
- **Average:** `double avg = (count > 0) ? (double)sum/count : 0.0;`
- **Min/Max:** `int m = A[0]; for(i=1;i<n;i++) if(A[i]>m) m = A[i];`

5. Dr. Gordon Table Tracing (40% of Exam!)

- **Columns:** Iteration, each variable (`i`, `x`, `sum`, array cells), and **Output**.
- **Rule:** Only write into a cell when its value **changes**.
- **Garbage Value (G / ?):** Uninitialized variables or array access out of bounds (e.g., `A[4]` for size 4) prints **Garbage (G)**!
- **Example Gordon Trace:** `int i, A[3]={1,2,3}; for(i=0;i<3;i++) cout<<A[i]; cout<<A[i];` → Loops print `1 2 3`, final `A[3]` is out-of-bounds → prints **Garbage**!

6. 1D & 2D Arrays & C-Strings

- `int A[5] = {10, 20};` → A[0]=10, A[1]=20, A[2]=A[3]=A[4]=**0**!
- **Indices:** 0 to n-1. `A[n]` is out-of-bounds error!
- **2D Matrix M[R][C]:** Rows `i` (0..R-1), Cols `j` (0..C-1).
- **Transpose:** Swap `M[i][j]` and `M[j][i]` for `j > i`.
- **C-Strings:** Null-terminated `char s[20];` → ends with `'\0'`. `strlen()` ignores null. `cin >> s;` stops at space; `cin.getline(s, 20);` reads full line.

7. Functions & Parameter Passing

- **Pass by Value:** `void f(int x)` → copy created, caller unchanged.
- **Pass by Reference:** `void f(int &x)` → alias, **modifies caller**!
- **Arrays to Functions:** `void f(int A[], int n)` always passes by reference (pointer); modifications inside change caller array!
- **Swap:** `void swap(int &a, int &b){ int t=a; a=b; b=t; }`

8. File I/O, Pointers & Checklist

- **Files:** `#include <fstream>` → `ifstream fin("in.txt"); if(!fin){ cout<<"error"; return 1; }` while(fin >> x){ ... } `ofstream fout("out.txt"); fout << ans; fout.close();`
- **Pointers & Dynamic:** `int *p = new int[n];` ... `delete[] p; p = nullptr;`
- **Math:** `#include <cmath>` → `sqrt(x)`, `pow(b,e)`, `fabs(d)`, `sin(rad)`
- **Exam Traps:** 1. Cast before divide · 2. `==` not `=` · 3. Loop trips count · 4. Check array bounds · 5. Tracing: write changes only!